

JKBOSE CLASS 10TH - MATHEMATICS

Time: 3 Hours

Series: Half-Yearly / Pre-Board Examination | Code: HY-10-MATH | Total Questions: 40
Max Marks: 80 Marks**GENERAL INSTRUCTIONS:** Section A (1m each), Section B (2m each), Section C (3m each), Section D (4m each).**SECTION - A : OBJECTIVE TYPE QUESTIONS**

(1 Mark Each)

1. Which of the following can end with the digit 0 for any natural number n ? [1 Mark]
- (a) $(2 \times 3)^4$
 - (b) $(2 \times 5)^4$
 - (c) $(2 \times 7)^4$
 - (d) $(2 \times 12)^4$
2. State the Fundamental Theorem of Arithmetic. [1 Mark]
3. If p is a prime number and p divides a^2 , then p divides a . (State True or False) [1 Mark]
4. The pair of linear equations $3x + 5y = 3$ and $6x + ky = 8$ do not have solutions, if k equals: [1 Mark]
- (a) 5
 - (b) 10
 - (c) $\neq 5$
 - (d) $\neq 10$
5. If one of the zeroes of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is: [1 Mark]
- (a) 10
 - (b) -10
 - (c) 5
 - (d) -5
6. The number of polynomials having zeroes as -2 and 5 is: [1 Mark]
- (a) 1
 - (b) 2
 - (c) 3
 - (d) More than 3
7. The zeroes of the quadratic polynomial $x^2 + px + p = 0$ ($p \neq 0$) are: [1 Mark]
- (a) Both equal
 - (b) Both cannot be positive
 - (c) Both unequal
 - (d) Both cannot be negative
8. The sum of the reciprocals of the roots of the quadratic equation $x^2 + px + q = 0$ is: [1 Mark]
- (a) p / q
 - (b) $-p / q$
 - (c) q / p
 - (d) $-q / p$

9. The roots of the quadratic equation $3x^2 - 4x + 3 = 0$ are: [1 Mark]
(a) Real and unequal
(b) Real and equal
(c) Imaginary (No real roots)
(d) None of these
10. For the quadratic equation $x^2 - 2x + 1 = 0$, the value of $x + 1/x$ is: [1 Mark]
(a) -1
(b) 1
(c) 2
(d) -2
11. The roots of the quadratic equation $x^2 - 2x - (r^2 - 1) = 0$ are: [1 Mark]
(a) $1 - r, r - 1$
(b) $1 - r, r + 1$
(c) $1, r$
(d) $1 - r, r$
12. The value of $(\sin 30^\circ + \cos 60^\circ) - (\sin 60^\circ + \cos 30^\circ)$ equals: [1 Mark]
(a) 0
(b) $1 + 2/\sqrt{3}$
(c) $1 - \sqrt{3}$
(d) $1 + \sqrt{3}$
13. If Triangle ABC is right-angled at C, then the value of $\cos(A + B)$ is: [1 Mark]
(a) 0
(b) 1
(c) $1/2$
(d) $\sqrt{3}/2$
14. Define the term 'Angle of Elevation' in heights and distances. [1 Mark]
15. Give one practical daily-life application of Trigonometry. [1 Mark]
16. Define a 'Sure Event' in probability theory. [1 Mark]
17. A bag contains 3 red balls and 5 green balls. If a ball is drawn at random, then the probability of getting a red ball only = [1 Mark]
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18. If two standard dice are thrown in the air, the probability of getting a sum of 3 will be [1 Mark]
19. A card is selected at random from a well-shuffled deck of 52 cards. The probability of being a face card is [1 Mark]
20. If an event cannot occur (impossible event), then its probability is [1 Mark]
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SECTION - B : VERY SHORT ANSWER QUESTIONS

(2 Marks Each)

21. Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$. [2 Marks]
22. Find the quadratic polynomial whose sum of zeroes is 4 and product of zeroes is 1. [2 Marks]
23. Solve the pair of linear equations: $2x + 3y = 11$ and $2x - 4y = -24$. [2 Marks]
24. Find the roots of the quadratic equation $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorisation. [2 Marks]
25. In a right triangle PQR (right-angled at Q) with $PQ = 12$ cm and $PR = 13$ cm, find the value of $(\tan P - \cot R)$. [2 Marks]
26. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the drawn ball is: (i) red, (ii) not red? [2 Marks]

SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

27. Prove that $\sqrt{2}$ is an irrational number. [3 Marks]
28. Find the zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients. [3 Marks]
29. Meena went to the bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. Meena received 25 notes in all. Find how many notes of Rs. 50 and Rs. 100 she received. [3 Marks]
30. Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots. [3 Marks]
31. Is it possible to design a rectangular park of perimeter 80 m and area 400 m^2 ? If so, find its length and breadth. [3 Marks]
32. Prove the fundamental trigonometric identity: $1 + \tan^2 A = \sec^2 A$. [3 Marks]
33. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 (equally likely outcomes). What is the probability that it will point at: (i) 8, (ii) an odd number, (iii) a number greater than 2, (iv) a number less than 9? [3 Marks]
34. Two dice (one blue and one grey) are thrown at the same time. Write down all the possible outcomes. What is the probability that the sum of the two numbers appearing on top of the dice is: (i) 8, (ii) 13, (iii) less than or equal to 12? [3 Marks]

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the original number. [4 Marks]
36. A train travels a distance of 480 km at a uniform speed. If the speed of the train had been 8 km/h less, then it would have taken 3 hours more [4 Marks]

to cover the same distance. Find the speed of the train.

37. Prove the trigonometric identity: $(\sin \theta - \cos \theta + 1) / (\sin \theta + \cos \theta - 1) = 1 / (\sec \theta - \tan \theta)$ [4 Marks]

38. Prove the trigonometric identity: $(\sin \theta - 2 \sin^3 \theta) / (2 \cos^3 \theta - \cos \theta) = \tan \theta$ [4 Marks]

39. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30 deg, which is approaching the foot of the tower with uniform speed. Six seconds later, the angle of depression of the car is found to be 60 deg. Find the time taken by the car to reach the foot of the tower from this point. [4 Marks]

40. From a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30 deg and 45 deg, respectively. If the bridge is at a height of 3 m from the banks, find the width of the river. [4 Marks]